



كلية الهندسة بشبرا
FACULTY OF ENGINEERING- SHOUBRA

Customer Segmentation By Clustering Algorithms

Kmean , Kmean ++, Hierarchical Cluster, DBSCAN , Fuzzy C Mean

ML Project | Supervised By Dr Mohamed Hussein

About the Problem

□ Retail businesses collect large amounts of transactional data, such as purchase frequency, recency, monetary value, product diversity, and return/cancellation behavior. However, without proper analysis, this data cannot be used effectively to understand customer behavior. The challenge is to segment customers based on their purchasing patterns to:

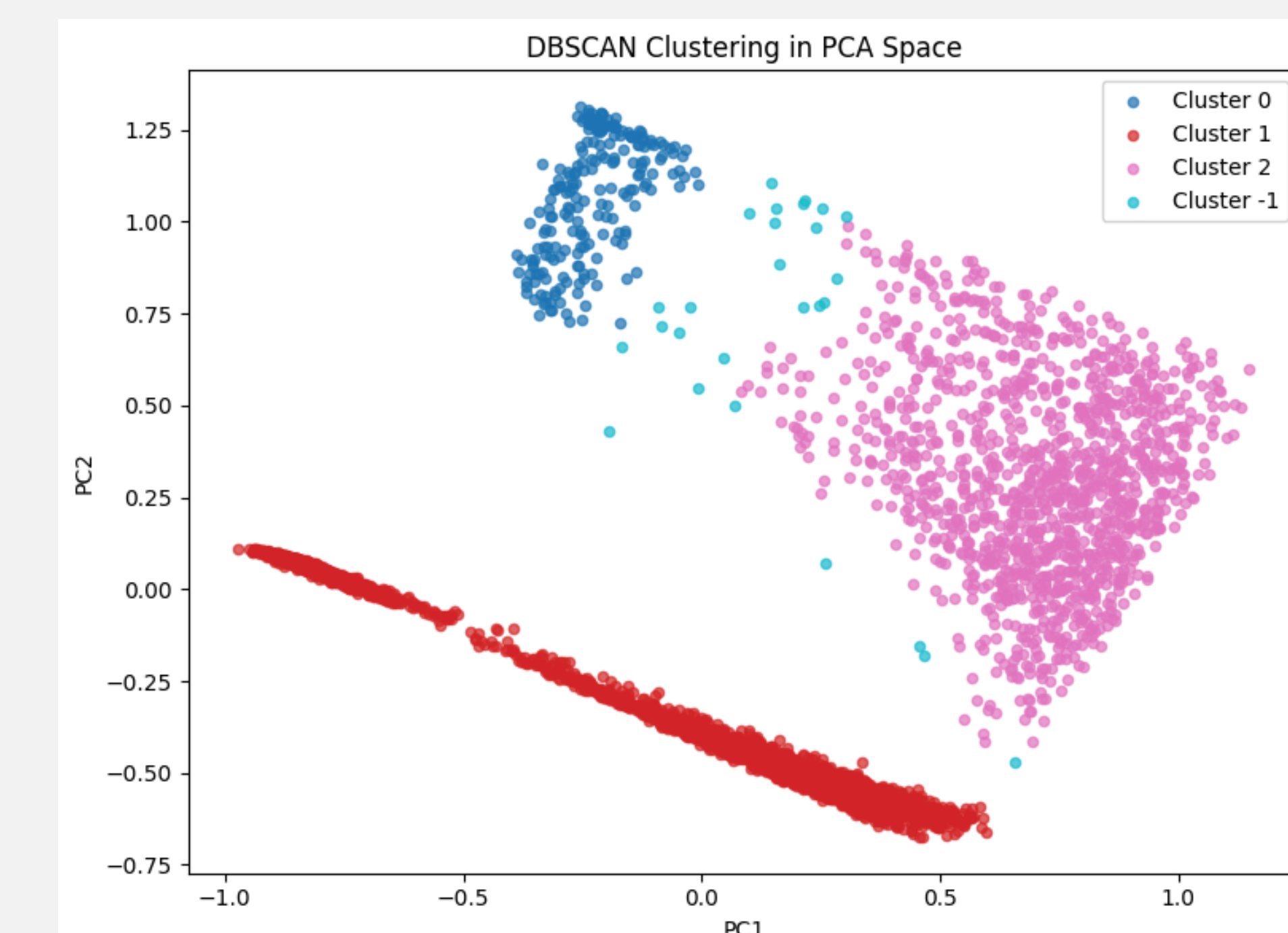
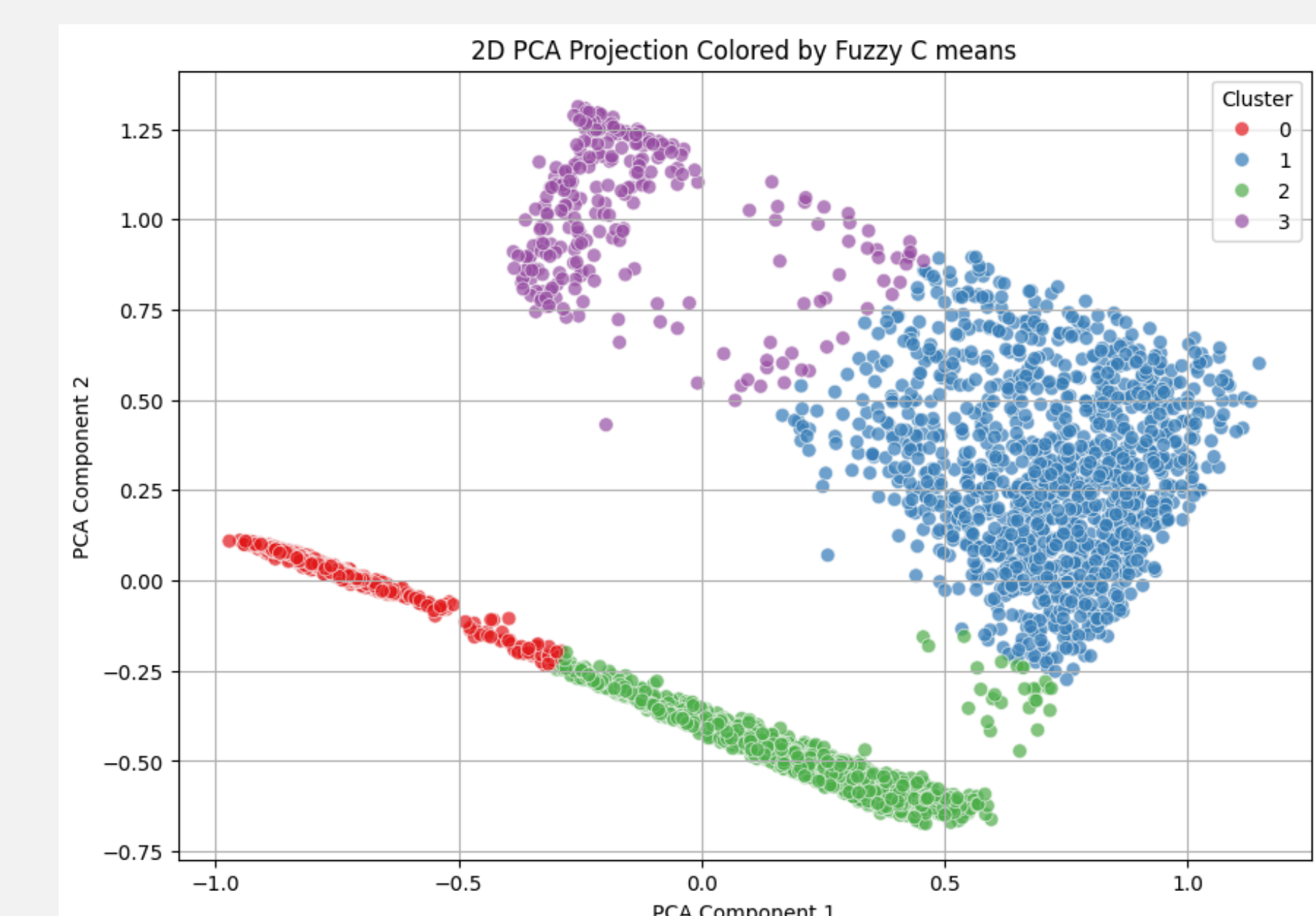
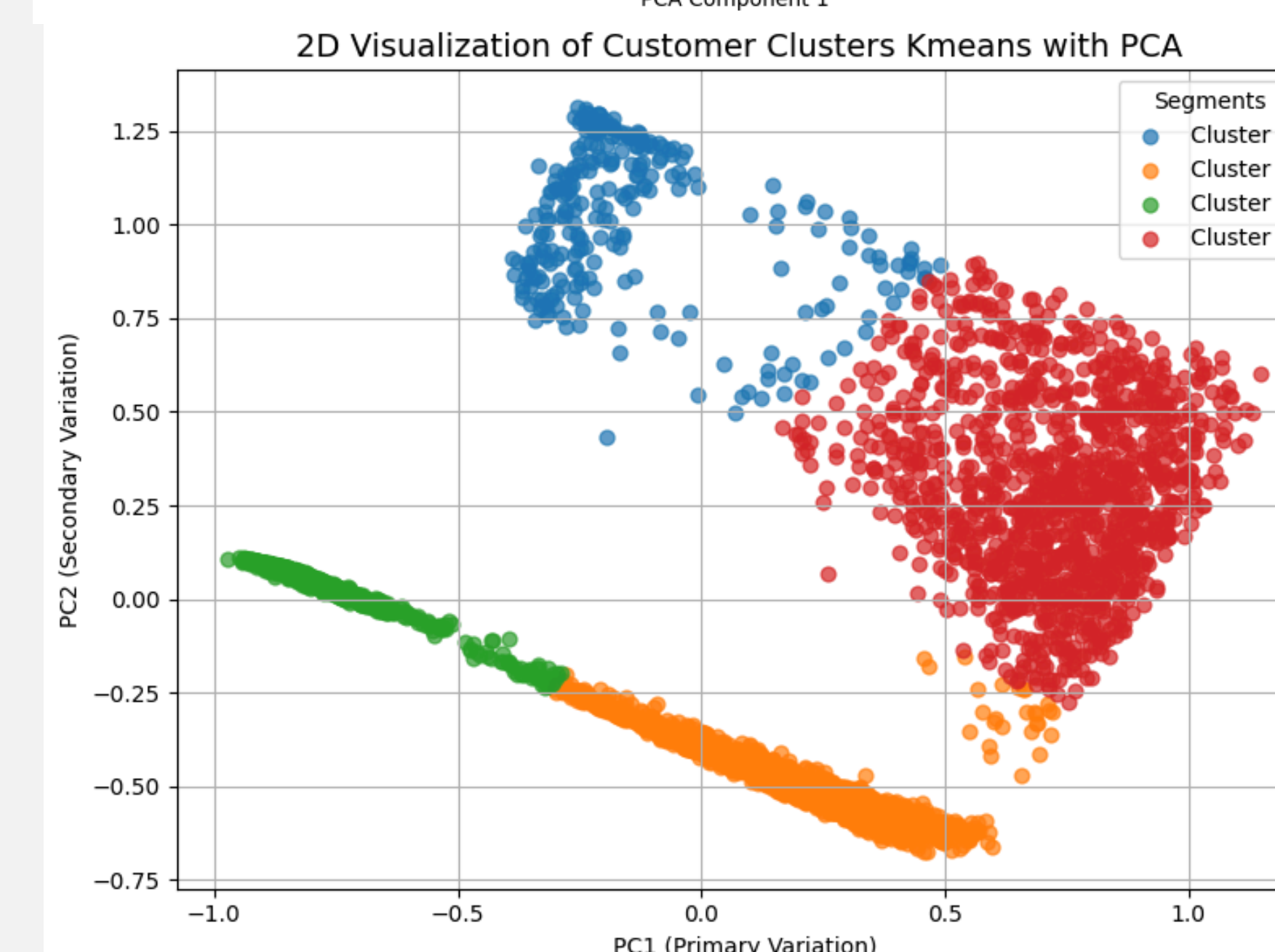
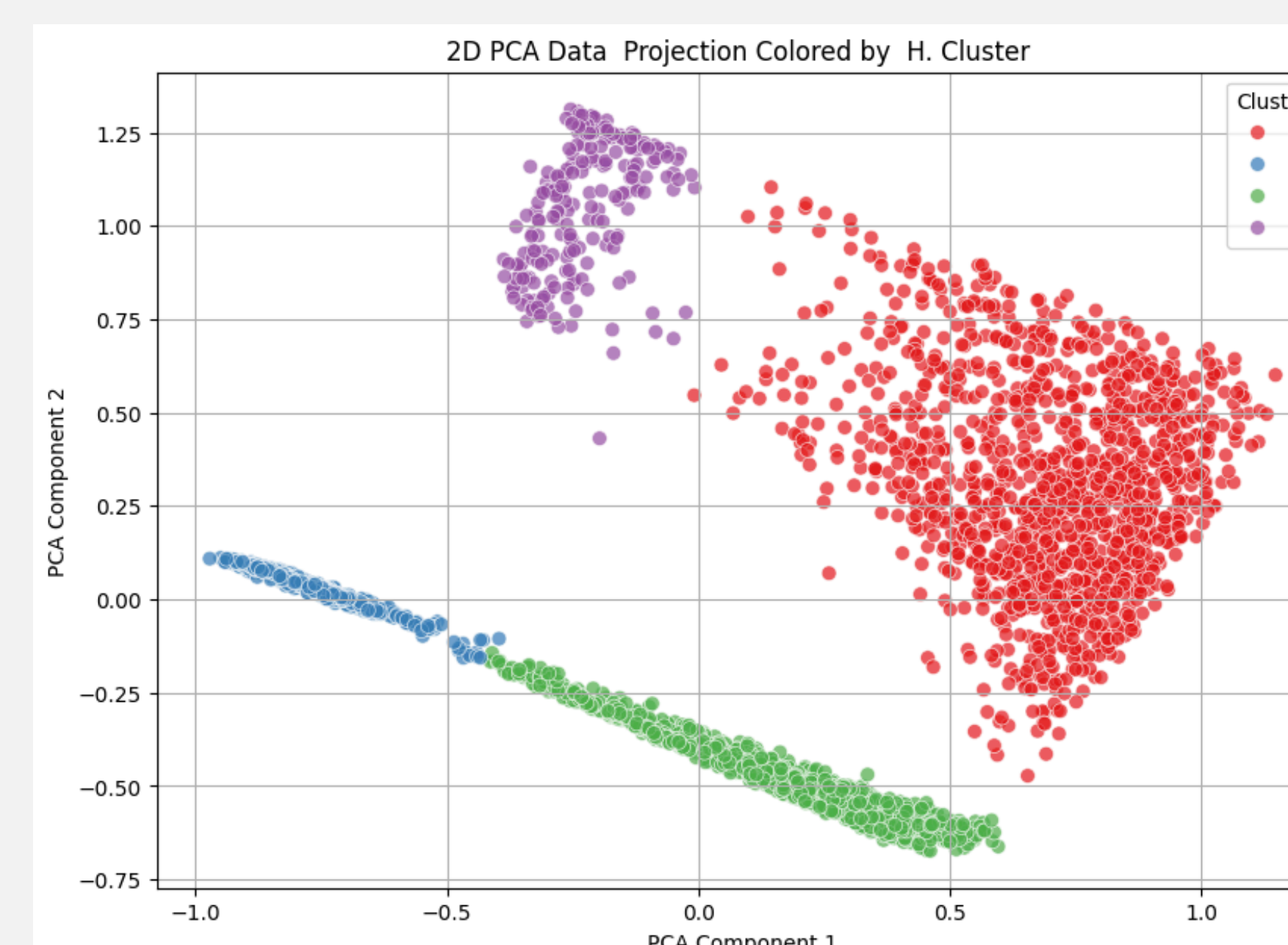
- Identify high-value and loyal customers for personalized marketing.
- Detect at-risk or one-time buyers for retention strategies.
- Understand behavior patterns to optimize products, promotions, and services.

Effective customer segmentation enables data-driven decisions, improves customer satisfaction, and increases revenue through targeted strategies.

Methodology

1. Data Collection and Importing
2. Exploratory Data Analysis (EDA)
3. Data Preprocessing & Feature Engineering
 - Created additional features: Return Rate, Tenure, Avg Days Between Purchases, Unique Products, Cancellation Rate.
4. RFM Analysis
5. Dimensionality Reduction
 - Applied PCA to reduce dimensionality and select key components for clustering.
6. Clustering Analysis
 - Applied K-Means Clustering on RFM and PCA data.
 - Applied Hierarchical Clustering
 - Applied Fuzzy C-Means Clustering on RFM data.
 - Applied DBSCAN Clustering on RFM & PCA Data
 - Evaluated clusters using silhouette scores and cluster profiling.
7. Cluster Profiling and Interpretation

Results



Cluster	Insight	Characteristics	Recommended Action
-1	Highly active but low value	Frequency very high, Monetary very low, Recency low, ReturnRate 12.5%, CancellationRate 32.5%	Monitor purchasing patterns, reduce returns/cancellations, educate customers
0	One-time buyers / trial customers	High Frequency, very low Monetary, Tenure = 1, CancellationRate 50%	Target with retention campaigns, incentives to increase value and loyalty
1	Stable core customers	Medium Frequency (107), medium Monetary (2.54), ReturnRate 0%, CancellationRate 0%, Avg_Days_Between_Purchases = 45	Maintain engagement, loyalty programs, cross-sell opportunities
2	High-value loyal / VIP customers	Low Frequency (44) but high Monetary (8.54), Tenure long (240 days), High Unique Products (106), low Return & Cancellation	VIP offers, personalized marketing, reward loyalty, upsell/cross-sell

Project Team : Aya Mohamed , Hanin Mustafa , Nadia Kamel